CEFALEXIN

Pregnancy: compatible

Cross-allergy with penicillin

CHEMICAL NAME (6R,7R)-7-[[(2R)-2-amino-2-phenylacetyl]amino]-3-methyl-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (monohydrate)

CLASS First-generation cephalosporin (oral)

TRADE NAMES Keflex · Ceporex · Cephalexin

THERAPEUTIC USES / INDICATIONS

- Skin and soft-tissue infections (streptococcal, MSSA).
- Urinary tract infections, including in pregnancy.
- Streptococcal pharyngitis (penicillin non-anaphylactic allergy).
- Bone infections (step-down oral therapy).

PHARMACODYNAMICS

- Binds **PBPs** → inhibits cell-wall synthesis → bactericidal.
- Good Gram-positive cover (streptococci, MSSA); limited Gram-negative (*E. coli*, *Klebsiella*, *Proteus*).

ADMINISTRATION

- **PO** capsules, tablets or suspension with or without food.
- Space doses evenly.
- Suspension: refrigerate after reconstitution; discard after 10–14 days.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, dyspepsia.
- Rash.
- Vaginal thrush.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (cross-reactivity with penicillin ~1–2%).
- *C. difficile* colitis.
- Interstitial nephritis; haemolytic anaemia (rare).

DOSAGES

- **Adult:** 250–500 mg every 6 h, or 500 mg every 8–12 h; max 4 g/day.
- **UTI (adult):** 500 mg every 12 h.
- **Children:** 25–50 mg/kg/day divided every 6–12 h (severe up to 100 mg/kg/day).
- Reduce in CrCl <30 mL/min.

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity.
- History of immediate (anaphylactic) penicillin allergy; avoid.
- **Caution:** renal impairment.

KEY INTERACTIONS

- **Metformin:** ↑ metformin levels.
- **Probenecid:** ↑ cefalexin levels.
- **Warfarin:** ↑ INR (minor).

PHARMACOKINETICS

- **Peak:** 1 h. Bioavailability ~90%.
- Not metabolised.
- Excreted unchanged in urine (high urinary levels). **t½:** 1 h.

NURSING MANAGEMENT

- Ask about penicillin/cephalosporin allergy and type of reaction.
- Monitor infection response and temperature.
- Monitor bowel habit.
- Check renal function in elderly.
- Collect urine culture in UTI before first dose.

PATIENT EDUCATION

- Complete the course.
- Report rash, swelling or breathing difficulty.
- Report severe diarrhoea.
- Shake suspension before use.

KEY POINTS

- Oral 1st-gen cephalosporin.
- Avoid after penicillin anaphylaxis.
- Skin infections and UTIs, safe in pregnancy.
- Renally excreted.